

DPEN021: Enabling Mathematics A

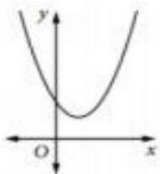
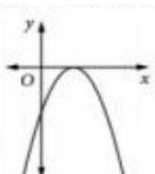
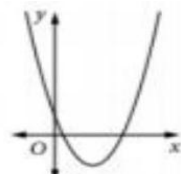
Class Questions Week 5

Week 5A

Key Idea - Composite Functions

- Given $f(x) = 3x - 5$ and $g(x) = x + 1$ evaluate:
(a) $f(g(x))$ (b) $g(f(x))$
- Given $f(z) = 2z$ and $g(x) = \frac{1}{x^2}$ evaluate:
(a) $f(g(x))$ (b) $g(f(z))$
- Given $f(x) = 4x^3 - x$, $g(t) = \frac{1-t}{1+t}$ and $h(z) = 2z$ evaluate:
(a) $g(f(x))$ (b) $f(g(t))$ (c) $f(h(z))$
(d) $h(f(z))$ (e) $g(h(t))$ (f) $h(g(x))$
- Given $f(x) = 1 - x^2$, $g(t) = 2t + 3$ and $h(y) = \frac{1}{y}$ evaluate:
(a) $g(f(x))$ (b) $f(h(y))$ (c) $h(g(t))$
(d) $h(f(-2))$ (e) $g(h(5))$ (f) $f(g(0.5))$

Key Idea - Parabolas and the Discriminant

- Calculate the discriminant for each of the following equations and state whether the equations have two, one or no real roots.
(a) $x^2 + 6x + 2 = 0$ (b) $2x^2 + 3x + 4 = 0$ (c) $4x^2 - 12x + 9 = 0$
(d) $-3x^2 + 2x - 1 = 0$ (e) $2x^2 = 3x + 7$ (f) $7x^2 = 63$
(g) $x(2x - 3) = 0$ (h) $(x + 6)^2 = x + 6$ (i) $(x - 1)(x - 2) + (x + 1)(x + 3) = 0$
- Which curve can you say that $\Delta < 0$.
(a)  (b)  (c) 
- Find the value for k for which the following equations have
(i) one real root (ii) two real roots (iii) no real roots
(a) $x^2 - 3x + k = 0$ (b) $(3k + 4)x^2 - 2x + 1 = 0$
(c) $-2kx^2 + 4x + 3 = 0$ (d) $x^2 + 2x + k - 1 = 0$
- Find the value for k for which the following equations have one root.
(a) $x^2 + (k - 1)x = 2k + 1$ (b) $x^2 + (k - 2)x + 1 = 0$

Week 5B

Key Idea - Quadratic Function

- Sketch the graph of the following curves indicating
 - x and y intercepts
 - axis of symmetry
 - vertex

(a) $y = x^2$	(b) $y = -x^2$	(c) $y = x^2 + 2$
(d) $y = x^2 - 9$	(e) $y = (x + 3)^2$	(f) $y = (x - 2)^2$
(g) $y = (x + 4)^2 - 1$	(h) $y = (x - 2)^2 + 3$	(i) $y = x(x - 2)$
(j) $y = (x + 1)(x - 3)$	(k) $x = -y^2 - 2y + 8$	(l) $x = 2y^2 - 6y + 7$
- Find the maximum or minimum values of the following equations:

(a) $f(x) = x^2 + 2x - 5$	(b) $f(x) = 7 - x - 3x^2$	(c) $f(x) = -2x^2 + 2x + 1$
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- The parabola $y = 2x^2 - kx + 4$ is symmetrical about the line $x = \frac{5}{4}$. Find the value of k and the minimum value.
- Find the domain and the range of the following:

(a) $y = x^2 + 2x + 1$	(b) $y = -x^2 + 3x - 2$	(c) $y = x^2 - 4x + 6$
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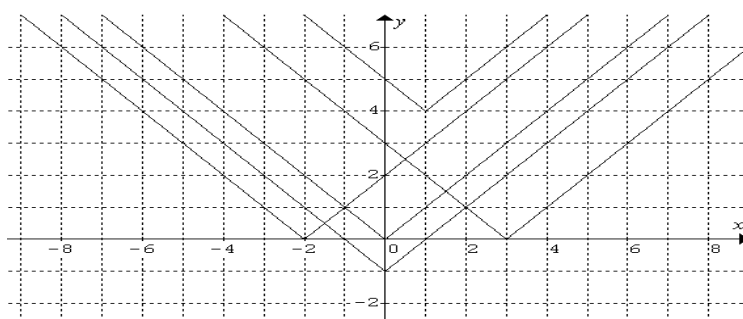
Week 5C

Key Idea - Absolute Value Function

- Draw the graph of the following, stating the x and y intercepts.

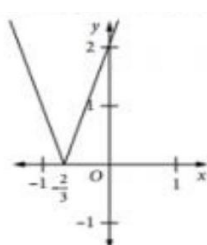
(a) $y = x + 2$	(b) $y = x - 3$	(c) $y = x + 4$
(d) $y = x - 1 $	(e) $y = x + 4 $	(f) $y = x - 3 $
(g) $y = x + 2 - 5$	(h) $y = x - 2 + 1$	(i) $y = x - 3 - 4$
(j) $y = 3x - 6 $	(k) $y = 4 - 2x $	(l) $y = 5x + 2 $
- Match each equation with its graph and give reasons for your choices.

(a) $y = x $	(b) $y = x - 1$	(c) $y = x - 3 $	(d) $y = x + 2 $	(e) $y = x - 1 + 4$
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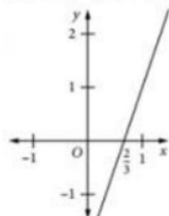


- Which diagram is the correct sketch of $y = |3x - 2|$?

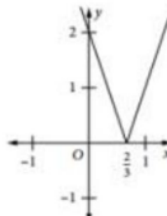
(a)



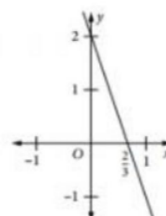
(b)



(c)



(d)



4. Find the domain and the range of the following:

(a) $y = |x| - 2$

(b) $y = |x - 1| + 5$

(c) $y = |2x + 3|$

Key Idea – Circles

1. Find the equation of each of the following circles.

(a) Centre: (3, 2)

(b) Centre: (-2, 1)

(c) Centre: (-3, -4)

Radius: 3

Radius: 2

Radius: $\sqrt{5}$

(d) Centre: (3, -3)

(e) Centre: (4, 0)

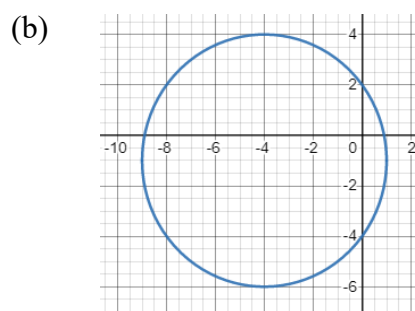
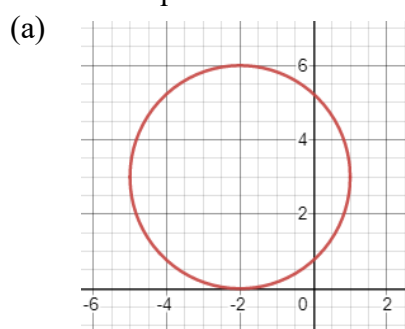
(f) Centre: $\left(-2, \frac{5}{2}\right)$

Radius: $\sqrt{7}$

Radius: 3

Radius: $\frac{\sqrt{13}}{2}$

2. Write the equation for each circle:



3. Find the coordinates of the centre and the radius for the following circles. Then sketch the circle. It is not essential to calculate the x and y intercepts for this question.

(a) $x^2 + y^2 - 6x + 4y - 3 = 0$ (b) $x^2 + y^2 + 4x + 2y - 4 = 0$ (c) $x^2 + y^2 - 2x + 4y = 0$

(d) $x^2 + y^2 - 6x + 6 = 0$ (e) $x^2 + y^2 + 4y - 5 = 0$ (f) $x^2 + y^2 + 3y - 5x = 1$

(g) $x^2 - 2x + y^2 + 3y = 2$ (h) $x^2 + y^2 - 8x + 5y = -3$ (i) $x^2 + y^2 + 9x - 4y = 24$

4. For the following equations find centre and radius of the circle. Then calculate the x and y intercepts and sketch the circle.

(a) $(x + 2)^2 + (y - 3)^2 = 9$ (b) $x^2 + y^2 + 2y = 5$ (c) $x^2 + y^2 + 8x + 2y = 8$

5. Show that the origin is a point on the circle $x^2 + y^2 + 6x - 8y = 0$.

6. Find the domain and the range of the following:

(a) $(x - 2)^2 + y^2 = 4$ (b) $(x - 1)^2 + (y + 5)^2 = 9$ (c) $(x - 3)^2 + (y - 5)^2 = 7$

Week 5D

Key Idea – Introduction to Logarithms

1. Evaluate the following without using a calculator.

(a) $\log_5 25$

(b) $\log_5 5$

(c) $\log_5 1$

(d) $\log_4 64$

(e) $\log_4 2$

(f) $\log_7 \frac{1}{7}$

(g) $\log_8 \frac{1}{64}$

(h) $\log_{10} 0.001$

(i) $\log_8 2$

(j) $\log_e \sqrt{e}$

(k) $\log_4 1$

(l) $\log_2 \frac{1}{8}$

2. Evaluate the following correct to 3 decimal places.
- | | | |
|---------------------|-------------------|-----------------|
| (a) $\log_{10} 400$ | (b) $\log_{10} 6$ | (c) $\log 0.25$ |
| (d) $\log_e 400$ | (e) $\log_e 6$ | (f) $\ln 0.25$ |

Key Idea – Logarithm Rules

1. Without using your calculator, simplify each of the following.
- | | | |
|---------------------------------|--|----------------------------------|
| (a) $\log_a a$ | (b) $\log_a a^3$ | (c) $\log_6 6^{2x}$ |
| (d) $\log_3 9$ | (e) $\log_3 27$ | (f) $\log_2 32$ |
| (g) $\log_5 \sqrt{5}$ | (h) $\log_e e^{x+1}$ | (i) $\ln e^{5x}$ |
| (j) $\log_3 \frac{1}{27}$ | (k) $\log_4 0.25$ | (l) $\log_9 3$ |
| (m) $-2 \log_2 8$ | (n) $\frac{1}{2} \log 10^{8x}$ | (o) $2 \ln e^{x+1}$ |
| (p) $\frac{\log a^6}{\log a^2}$ | (q) $3 \log_2 \frac{1}{16}$ | (r) $\frac{\log_2 16}{\log_2 2}$ |
| (s) $e^{\ln x}$ | (t) $e^{-\ln\left(\frac{1}{x}\right)}$ | (u) $e^{x+\ln x}$ |
2. Without using your calculator, simplify each of the following.
- | | | |
|--|---|-------------------------------------|
| (a) $\log_{10} 2 + \log_{10} 5$ | (b) $\log_6 4 + \log_6 9$ | (c) $\log_a 5 + \log_a \frac{1}{5}$ |
| (d) $\log_{10} 25 + \log_{10} 40$ | (e) $\log_2 16 + \log_2 8$ | (f) $\log_4 128 - \log_4 2$ |
| (g) $\log_3 54 - \log_3 18$ | (h) $\log_2 20 - \log_2 5$ | (i) $\log_2 20 - \log_2 10$ |
| (j) $\log_2 1 - \log_2 2$ | (k) $\log_3 4 - \log_3 36$ | (l) $\log_3 54 - \log_3 18$ |
| (m) $\log_2 18 - 2 \log_2 3$ | (n) $\frac{1}{2} \log_{10} 16 + 2 \log_{10} 5$ | (o) $(\log_2 16)(\log_2 4)$ |
| (p) $\log_{10} 50 + \log_{10} 8 - \log_{10} 4$ | (q) $\log_{10} 125 + \log_{10} 32 - \log_{10} 4$ | |
| (r) $\frac{1}{3} \log_{10} 8 + 3 \log_{10} 2 - \log_{10} 16$ | (s) $\log_{10} (3x^2) - \log_{10} (x^2)$ | |
| (t) $\log_e 4x - \log_e 2x$ | (u) $2 \log_{10} 5x + 3 \log_{10} x - \log_{10} 25$ | |
3. Use the change of base rule to evaluate the following correct to 3 decimal places:
- | | | |
|--------------------------|------------------|------------------|
| (a) $\log_8 80$ | (b) $\log_2 10$ | (c) $\log_5 100$ |
| (d) $\log_5 \frac{1}{3}$ | (e) $\log_4 0.3$ | (f) $\log_3 2$ |

Key Idea – Solving Equations with Logarithms

1. Solve the following equations, give your answer in exact form.
- | | | |
|------------------------|--------------------|----------------------------|
| (a) $\log_6 x = 3$ | (b) $\log_2 x = 5$ | (c) $\log x = 0.6$ |
| (d) $6 \log(2x+5) = 3$ | (e) $\ln(5x) = 3$ | (f) $5 \ln(3x-1) + 3 = 23$ |
2. Solve the following equations, give your answer correct to 3 decimal places.
- | | | |
|----------------------|---------------------------|------------------------------|
| (a) $\log_3 x = 10$ | (b) $\log_3 x = 3.2$ | (c) $\ln(3x) = 2$ |
| (d) $3 \log(4x) = 6$ | (e) $3 \log_e(4x-1) = 18$ | (f) $2 \log_e(2x+1) + 1 = 7$ |

3. Solve the following equations, give your answer in exact form.

(a) $5 \ln x - 3 = 2 \ln x$

(b) $\log_{10} x^2 = 2$

(c) $2 \log_{10} x + 3 = \log_{10} (x^5)$

(d) $\log_{10} x^4 + 1 = 3 \log_{10} x - 1$

(e) $\ln x = \ln 5 + \ln 3$

(f) $\log_{10} x = \log_{10} 4 + \log_{10} 2$

(g) $\log x = 4 \log 2 + 2 \log x$

(h) $\log_{10} x = \frac{1}{2} \log_{10} \left(\frac{1}{4} \right)$

(i) $3 \log_{10} 2 + \log_{10} 3 = \log_{10} x$

(j) $\log_2 16 + \log_2 x = \log_2 32$

(k) $\log_{10} x - \log_{10} (x-1) = 1$

(l) $\log_{10} 5 + \log_{10} x - \log_{10} 3 = 2$